

Abdullah Usama

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Education

National University of Sciences and Technology (NUST), Islamabad

Bachelor of Software Engineering, SEECS

Sept 2022 – Sept 2026

- **Coursework:** Data Structures and Algorithms, Object Oriented Programming, Web Engineering, Operating Systems, Database Systems, Design and Analysis of Algorithms, Machine Learning, Reinforcement Learning, Embedded Systems, Large Language Models

Experience

AI Engineer - RapidsAI, Islamabad

October 2025 – Present

- Building production-grade **AI Agents** for workflow automation, data processing, decision support, and multi-step task orchestration.
- Designing and implementing **agentic system architectures**, including tool-using agents, retrieval-augmented reasoning loops, and multi-agent collaboration frameworks.
- Developing automation pipelines using **LangChain**, **LangGraph** and **n8n**, integrating LLMs with external APIs, databases, and custom tools to streamline business processes.

ML Intern - OneScreen Solutions, San Diego, California – Remote

April 2025 – June 2025

- Worked with Vision Transformers (ViT-32) and Vision-Language Models (VLMs) like PaLI-Gemma to explore multimodal learning and representation.
- Developed an end-to-end pipeline combining **YOLO** for object detection and **SAM** for segmentation to remove occlusions (e.g., people) from lecture videos, reconstructing clean full-frame images.
- Optimized dataset quality and object localization by merging SAM's pixel-level masks with YOLO annotations, reducing label noise and improving mean Average Precision (mAP) by **7–10%**.

Freelance AI Developer

March 2025 – Present

- Developed and deployed custom Retrieval-Augmented Generation (RAG) based conversational AI chatbots, enabling natural and context-aware interactions.
- Engineered a comprehensive administrative dashboard for clients to manage chatbot knowledge bases, including adding new documents and monitoring user interactions.
- Designed and implemented AI-powered Learning Management Systems (LMS) with features tailored for personalized learning experiences and automated content delivery.

Computer Vision Intern - Machine Vision & Intelligent Systems Lab (MachVis), SEECS

June 2024 – Aug 2024

- Engineered and optimized computer vision pipelines for real-time object detection, tracking, and motion estimation in dynamic environments.
- Designed and implemented robust feature extraction methods, including ORB and optical flow algorithms, to estimate camera motion and stabilize frame sequences.
- Developed and optimized real-time tracking systems using SORT and Kalman Filters to maintain consistent object identities across video frames.

Projects

MACHINE LEARNING, COMPUTER VISION AND AI

Multimodal RAG AI Chatbot

Designed and implemented a high-performance multimodal Retrieval-Augmented Generation (RAG) chatbot capable of processing PDFs and images, supporting conversational queries, and handling high-concurrency workloads with an optimized async architecture.

- Built a **multimodal RAG pipeline** using **ColPali / Qwen3-VL embeddings** for joint text-image representation, storing vectors and metadata in **Qdrant**, with original documents hosted on **Cloudinary**.
- Engineered both **synchronous and asynchronous APIs** using **FastAPI**, introducing a task-based system that returns **task_id** in $\sim 40\text{ms}$ and enables real-time progress tracking for long-running indexing and query jobs.
- Implemented a scalable **distributed task queue architecture** using **Celery + RabbitMQ** with **Redis** as a result backend, enabling efficient handling of high-concurrency user requests.
- Engineered an optimized query pipeline combining **conversational memory + query rewriting (Gemini 2.5 Flash)**, an intelligent **query parser with early-exit optimization** to skip unnecessary RAG steps, and a **vision-based reranking stage** to filter irrelevant images, significantly improving relevance, reducing hallucinations, and lowering latency and compute cost.
- Implemented configurable controls (**top_k, temperature, image_quality**) for tuning performance and output quality, alongside robust **document management APIs** (listing, deletion) and optimized image retrieval via thumbnail storage in the vector database.

AI-Powered Competitive Intelligence Automation (n8n)

Developed an end-to-end **n8n-based automation pipeline** to replace manual competitive analysis workflows by integrating Google Sheets data, AI-driven scoring, and strategic insights generation.

- Designed multi-stage **AI workflows** to process large-scale data (3,000+ rows) from Google Sheets, performing **sentiment analysis, competitive impact scoring (0–100)**, and generating actionable insights across reviews, social media, website changes, and news data.
- Engineered **prompt-driven LLM pipelines** (GPT-4o, Gemini) with structured JSON outputs and optimized prompts for accurate scoring, while automating **competitor intelligence extraction** via AI-assisted search.

Skills: n8n, Prompt Engineering, Workflow Automation, Google Sheets API, Data Processing, Sentiment Analysis.

News AI Agent ([GitHub](#))

Developed an intelligent web-based assistant to help students and CSS/PMS aspirants understand and analyze DAWN - Op & Ed, The Tribune, ParadigmShift - National & IR by offering topic-wise batch search, vocabulary insights, idioms, summaries, and article URLs through an AI-powered backend and interactive frontend.

- Integrated **tool calling via LangChain**, enabling the LLM to autonomously invoke backend tools (e.g., search, summarization, and vocabulary analysis functions) based on user intent. This architecture uses LangChain's **Tool** and **AgentExecutor** modules to dynamically route model queries to Python-defined functions, improving reasoning control, modularity, and reliability of task execution.
- Built a **FastAPI-based backend** integrated with **LangChain** and deployed on Render, featuring **article scraping, content extraction, and flexible date parsing** for natural language queries powered by **Gemini 2.0 Flash**.

Skills: FastAPI, LangChain, Google Gemini 2.0 Flash, Next.js, Web Scraping, REST APIs, Natural Language Processing, Vocabulary Analysis, Deployment (Render, Vercel), CORS, Date Parsing.

Mistral-7b-Instruct Finetuning (LLM Fine-tuning) (75+ Downloads on HuggingFace)

([Hugging Face](#)) ([GitHub](#))

June 2025

Fine-tuned Mistral-7B-Instruct-v0.3 using LoRA on a custom dataset of opinion articles by Pakistani diplomat and journalist Maleeha Lodhi to imitate her writing style.

- Scraped and curated **6 years of Dawn opinion articles**, then **cleaned, structured, and formatted** them into an instruction-response (JSONL) dataset with a tailored **prompt template** for fine-tuning Maleeha Lodhi's writing style.
- Conducted **Supervised and Parameter-Efficient Fine-Tuning (SFT + LoRA/PEFT)** under resource-constrained conditions using the **BitsAndBytes** library for quantized training, optimizing GPU memory usage while maintaining stylistic accuracy.

Skills: Large Language Models, Mistral-7B, LoRA, PEFT, SFT, Data Scraping, JSONL Formatting, Prompt Engineering, NLP Fine-Tuning

Pakistan Penal Code RAG Chatbot ([GitHub](#))

Developed a **Retrieval-Augmented Generation (RAG)** chatbot for the Pakistan Penal Code.

- Parsed, cleaned, and curated legal text data before applying a **recursive chunking pipeline**. Designed a **custom retriever with reranking** to prioritize high-relevance results and implemented **query parsing and reformulation** for improved matching and retrieval accuracy from the vector database.
- Developed a robust **Retrieval-Augmented Generation (RAG)** system using **Weaviate** with hybrid semantic-keyword search, powered by **Cohere's multilingual embeddings** and **Gemini AI**, and enhanced through a **hybrid recursive chunking strategy** for precise legal text retrieval.

Skills: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Vector Databases, Streamlit, Cohere, Google Gemini, Python, Natural Language Processing (NLP), Information Retrieval

Bounding Box Refinement Pipeline ([GitHub](#)) ([My Article](#))

Apr 2025 – May 2025

Built a SAM-powered preprocessing pipeline to automatically refine YOLO bounding boxes for enhanced dataset accuracy and model performance.

- Integrated **Segment Anything Model (SAM)** with YOLO annotations to tighten and correct bounding boxes, improving object localization and dataset quality while **boosting mAP by 7–10%** and **cutting re-annotation costs**.

Skills: Object Detection, YOLO, Segment Anything Model (SAM), Bounding Box Optimization, Data Preprocessing, Computer Vision, Model Evaluation

WEB DEVELOPMENT

Khappa POS – Point of Sale & Inventory Management System ([GitHub](#))

Developed a modern POS and inventory management solution with dual interfaces for cashiers and administrators, offering real-time sales tracking, inventory analytics, and data redundancy through Supabase and Google Sheets integration.

- Developed a robust **Next.js full-stack POS system** with a dual-panel design—Cashier Panel for smooth transactions and Admin Panel for inventory, reporting, and role-based management—powered by **Supabase (PostgreSQL & Auth)** for secure, scalable operations.
- Built real-time **sales analytics and backup workflows**, integrating **Chart.js** for revenue and performance insights and implementing a live **Google Sheets backup pipeline** to log all transactions for transparency and data safety.

Skills: Next.js, React.js, Supabase (PostgreSQL & Auth), Chart.js, Tailwind CSS, Google Sheets API, Responsive UI/UX, Data Visualization, Role-Based Access Control, POS Systems.

Plant E-Commerce App (MERN) ([GitHub](#))

Dec 2024 – Jan 2025

Developed a plant e-commerce platform with secure authentication, multilingual support, and seamless payment integration, enhancing user experience and accessibility.

- Developed a full-stack e-commerce platform with a **responsive React.js frontend** and a **scalable Node.js backend**, integrating **secure role-based authentication** with Clerk, **multi-language support**, and **Stripe-based payment processing** to provide a robust, user-friendly, and globally accessible shopping experience.

Skills: MERN Stack, Authentication & Authorization, Payment Gateway Integration, Multilingual Support, React.js, Node.js, Express.js, MongoDB, Clerk.com, Stripe, UI/UX Design, RESTful APIs.

Skills

- **Programming:** JavaScript, Python, C++, SQL
- **Frameworks & Libraries:** React.js, Next.js, Node.js, Express.js, OpenCV, MediaPipe, TensorFlow, LangChain, LangGraph, LangSmith
- **Databases:** MongoDB, MySQL, PostgreSQL
- **Cloud & Deployment:** Docker, RESTful APIs, Vercel, Render
- **Tools:** n8n, Git, GitHub, Clerk.com, Stripe, Pycaw, React-i18n